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Zeal College of Engineering & Research, Pune

Department of Electronics and Computer Engg

Subject : Basic Electronics Engg

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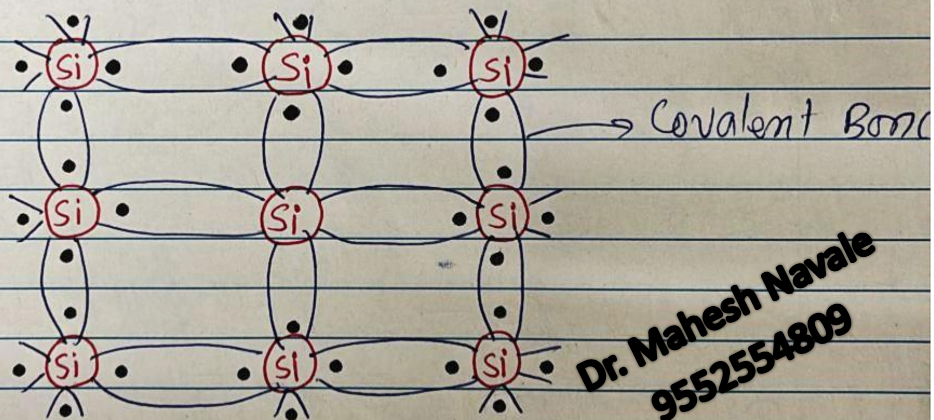
Unit II : Semiconductor Materials

- Semiconductors: P-type and N-type, Current in semiconductors: Diffusion and Drift Current.
- P-N Junction Diode: Construction, working in forward and reverse bias, V-I characteristics, Diode applications: Diode as a switch, Diode as Rectifier: HWR, FWR, BR, Specifications of Rectifier diodes.
- Special purpose diodes: Zener diode: V-I Charateristics, Specification and Zener as voltage regulator, Light Emitting Diode (LED) and photo diode.

Intrinsic Semiconductor :

- Pure form of semiconductor
- current flows due to exitation of electrons
- Reason behind this is temperature, heat, light etc.
- Very less free electrons ∴ ct. is less, called as "leakage ct"
- Comparatively Silicon is having less leakage ct. than Germanium
- Fig shows ~~Atl~~ Atomic Structure of Instrinsic S/C



- Here covalent bond is formed between two ~~silicons~~ Silicon atom itself.
- To balance, it must consist of 8 electrons in its outermost orbit
- Thus due to covalent bond, now each silicon consists of 8 electrons in its outer most orbit & becomes balance
- When any external energy such as temp,

light & etc is applied, these electrons becomes free due to which current flows through it.

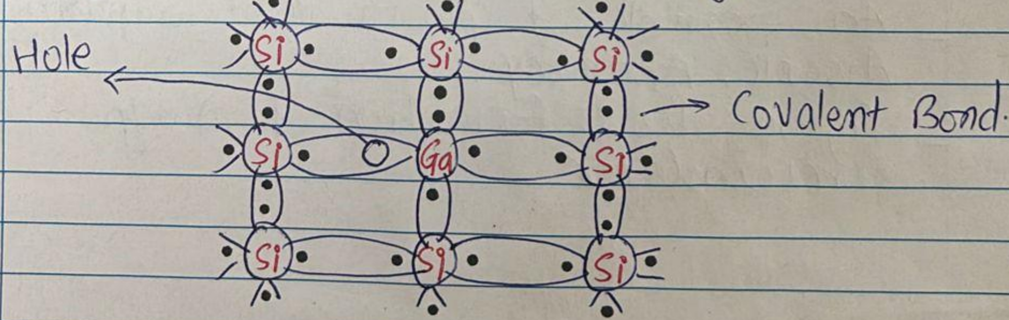
- furthermore if an external source is applied, due to ~~moment~~ movement of electrons, current flows through semiconductor.

* Extrinsic Semiconductor

- It is impure form of semiconductor
- It is formed by adding impurities to pure semiconductor
- The process of adding impurity is called as "Doping"
- Based on impurity added, there are two types of extrinsic semiconductor
  - P type -//-
  - N type -//-

(1) P Type Semiconductor :

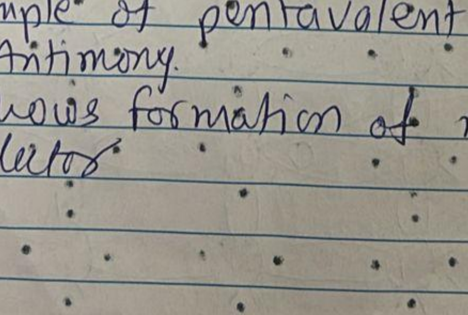
- Here a trivalent impurity is added with pure semiconductor
- These trivalent impurities are also called as "Acceptor Impurities"
- An ~~expa~~ example of trivalent impurity is - Gallium, Indium & etc.
- fig shows formation of P-type s/c

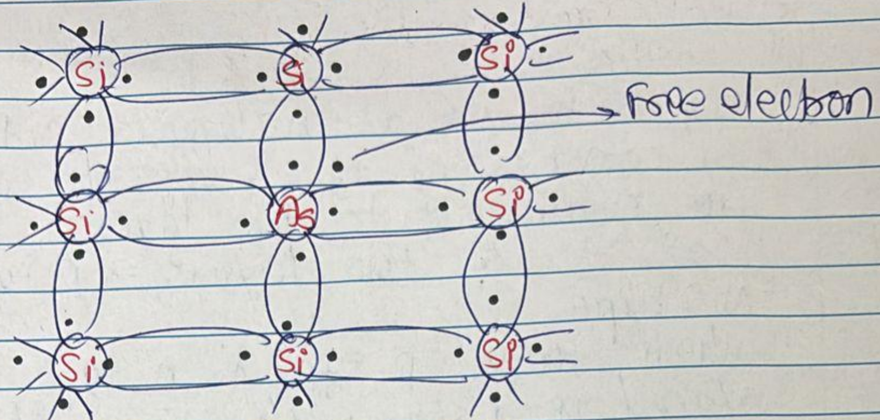


- Due to addition of trivalent impurity, large no. of holes are created in semiconductor
- Here covalent bond is formed between pure semiconductor & an impurity
- Thus it will be balanced due to addition of hole
- Where hole is a positive charged carrier
- And it represents a missing of electron
- As trivalent impurity has only 3 electrons in its outermost orbit, covalent bond is formed with the help of hole.
- In P type semiconductor, holes are majority carriers whereas electrons are minority carriers

2) N-Type Semiconductor

- Here a pentavalent impurities are added with pure semiconductor.
- These pentavalent impurities are also called as " Donar impurities
- An example of pentavalent impurity is Arsenic, Antimony.
- figure shows formation of n-type semiconductor



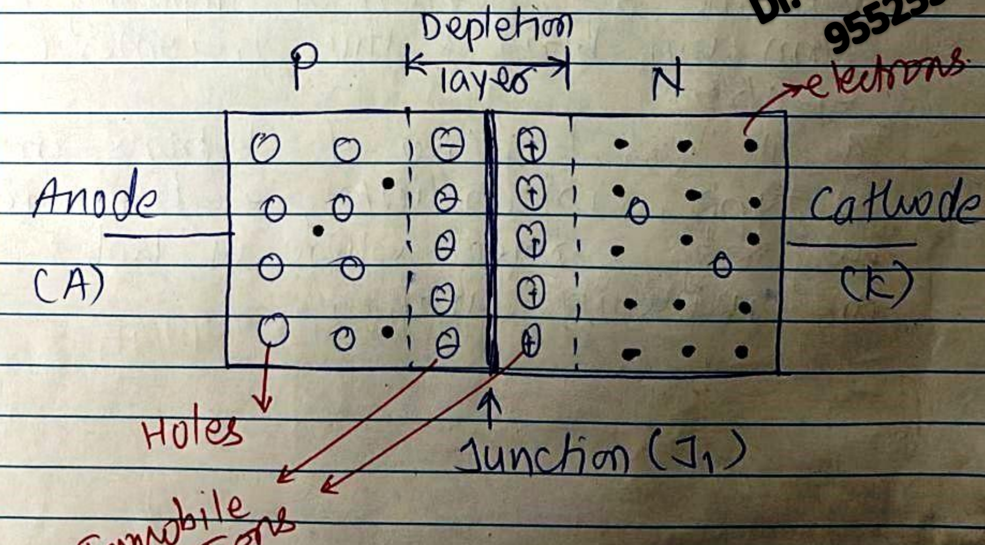


- Due to addition of pentavalent impurity, large no. of free electrons are created.
- As pentavalent impurity has 05 electrons in its outermost orbit, 04 electron of them forms a covalent bond with 04 electrons of pure semiconductor and becomes balance.
- But the 5th electron of impurity don't form any bond & thus becomes free electron.
- As it has large no. of electrons, in N type semiconductor, electrons are majority carriers where as holes are minority carriers.

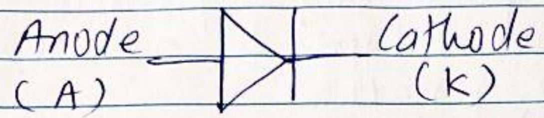
* PN Junction Diode

1) Construction / Structure :

- Diode is an abbrivation of Di-Electrode
- It consists of two electrodes which are connected to two layers.
- Thus it has two layers – P type & N-type
- When these p type & n type s/c placed near to each other, it forms a junction
- Thus diode is a Unijunction device
- when p type & n type s/c placed together, at the junction due to repulsion of majority carriers, a depletion layer is formed.
- figure shows structure of P-N junction diode



- The symbol PN junction diode is —

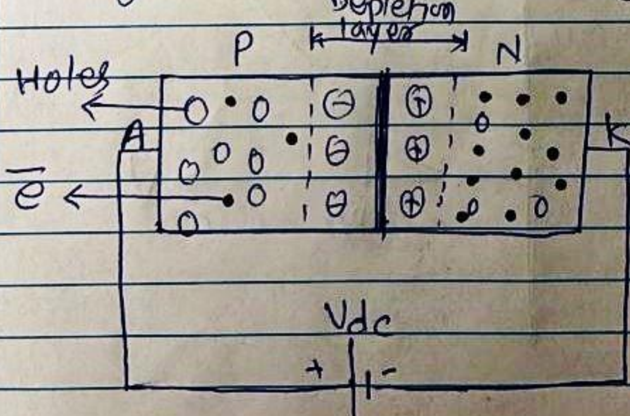


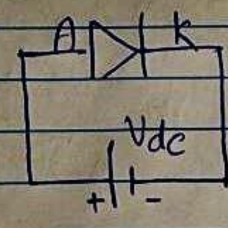
2) Working Principle / V-I characteristics :⇒

- Diode basically works in two different conditions
    1) Forward Bias Condition
    2) Reverse Bias condition

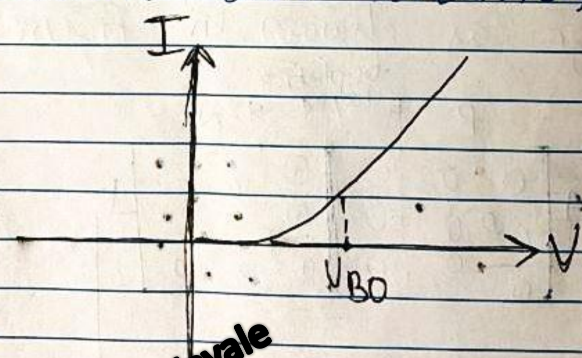
1) Forward Bias condition

- Here P type material (Anode) is connected to positive plate of supply voltage & N type material (Cathode) is connected to negative plate of supply voltage as shown in figure



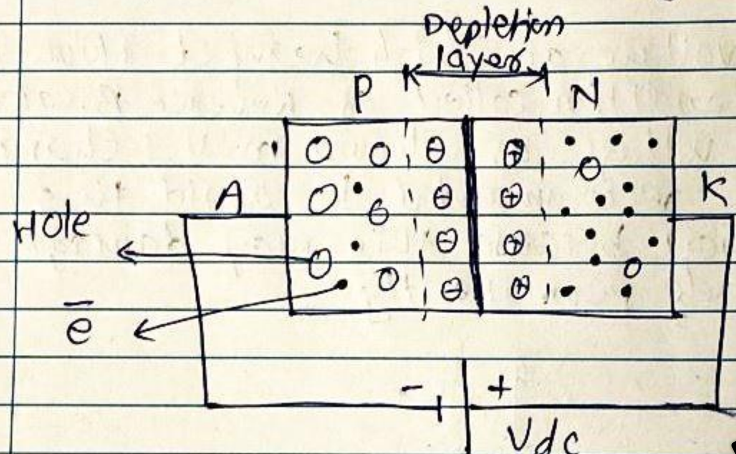


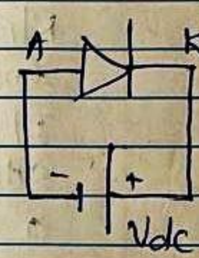
- When external DC voltage is applied, due to +ve plate, majority carriers (holes) from P type gets repelled towards the junction.
- Similarly due to -ve plate of the supply voltage, electrons from N type gets repelled towards junction
- Due to this, the junction barrier decreases
- When the supply voltage is further increased, one state will come at that time the junction barrier collapse & heavy current flows through diode
- This current is called as forward ct.
- And the voltage at which junction breaks, called as Break Over Voltage.
- Figure shows V-I characteristics in forward bias condition



2) Reverse Bias Condition

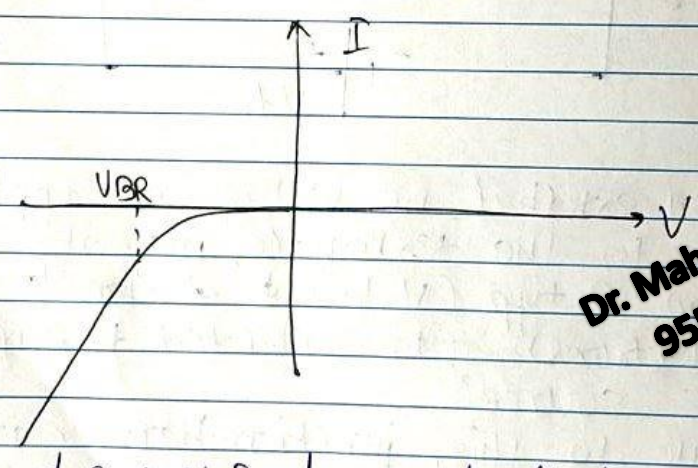
- Here P type material (Anode) is connected to -ve plate of supply voltage & N type material (Cathode) is connected to +ve plate as shown in figure



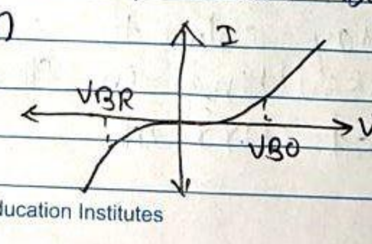


- When external DC voltage is applied due to the terminals, majority carriers from P type (Hole) & ~~electrons~~ & N type (electrons) gets attracted towards DC source
- Due to this, junction barrier increases
- At the same time, minority carriers may able to cross the junction, resulting in flow of small amount of current.

- This current is called as leakage current.
- When the reverse supply voltage is further increased, on existing covalent bond will break creating large no. of minority carriers.
- Due to this heavy current flows through diode
- The voltage at which heavy ct. flows in R/B condition called as Reverse Break-down voltage as shown in V-I chara.
- It is reccommanded to avoid this situation because this may damage a diode permanently.



- Fig. shows V-I chara of diode both in F/W & R/B condition



* Applications of Diode

1) Diode as a Rectifier

- Rectifier is basically used to convert bidirectional i/p signal (AC) to unidirectional o/p signal (DC)
- Based on the output, there are basic two types of Rectifiers.

- Half wave Rectifier
- Full wave Rectifier
  - Full wave Centre Tap Rectifier
  - Full wave Bridge Rectifier

* Half wave Rectifier

- Here only one diode is connected in series with load resistance (R_L) as shown in figure 1

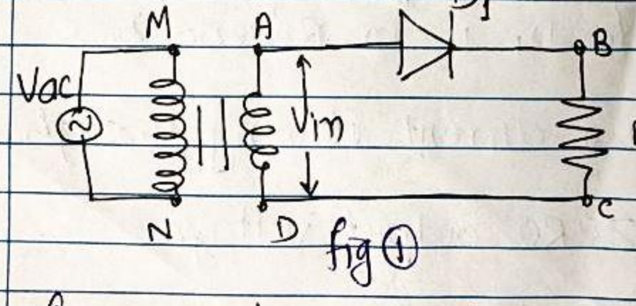


fig ①

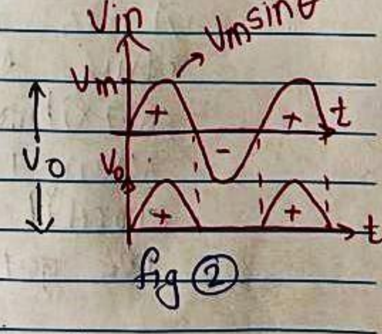


fig ②

- figure 2 shows input, output waveforms.

The working of the circuit is explained with the help of two different conditions of applied i/p cycle.

1) For Positive Half Cycle.

- M is positive with respect to N
- A is positive with respect to D
- Thus diode D_1 is in forward bias condition
- Current flows through the circuit as –
  $A - D_1 - B - R_L - C - D$
- Thus current flowing through R_L is from B to C ($R_L \downarrow$)
- Let us consider that this current produces positive o/p voltage. ∴ $\boxed{V_0 = +ve}$

2) For Negative Half Cycle.

- M is negative & N is positive
- A is negative & D is positive.
- Thus diode D_1 is in Reverse bias condition.
- There is no current flowing through circuit
- We get ZERO output voltage.

- Thus only for +ve Half cycle we get o/p ∴ HWR.

* Centre Tap Full wave Rectifier.

- Here a transformer which consists of a centre tapping at its secondary winding is used.
- Here two diodes are used as shown in figure 1

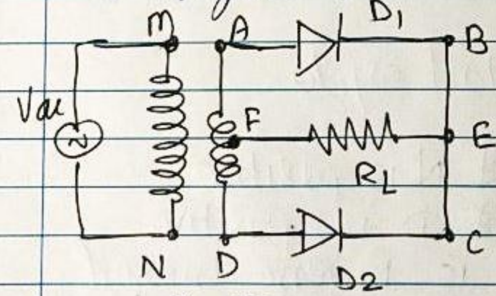


fig ①

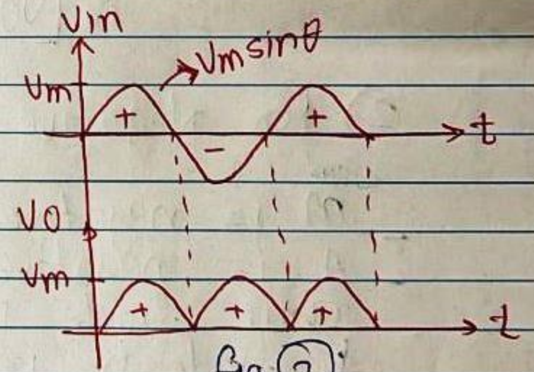


fig ②

- figure 2 shows input, output waveforms.

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* Working

1) For Positive Half Cycle

- M is positive w.r.to N.
- A is positive w.r.to D.
- Thus diode D_1 is forward biased & D_2 is Reverse biased.
- Current flows through the circuit as -
  $A - D_1 - B - E - R_L - F$
- Thus current flowing through R_L is from E to F ($\overleftarrow{R_L}$)
- Let us consider that this current produces positive o/p voltage

$\therefore$ $\boxed{V_0 = +ve}$

2) For Negative Half cycle

- M is negative & N is positive
- A is negative & D is positive
- Thus diode D_1 is Reverse biased & diode D_2 is forward biased.
- Current flows through the circuit as -
  $D - D_2 - C - E - R_L - F$
- Thus current flowing through R_L is from E to F ($\overleftarrow{R_L}$)

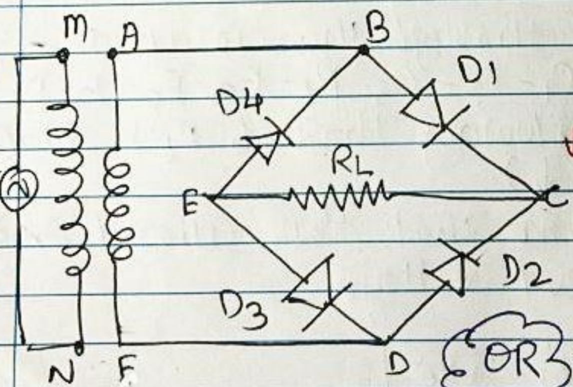
- And as we already assumed that if current flowing through R_L is from E to F, positive o/p voltage is produced

$\therefore \boxed{V_0 = +ve}$

- Thus, here both the half cycles of applied i/p, generates +ve o/p $\therefore$ it is called as FWR.

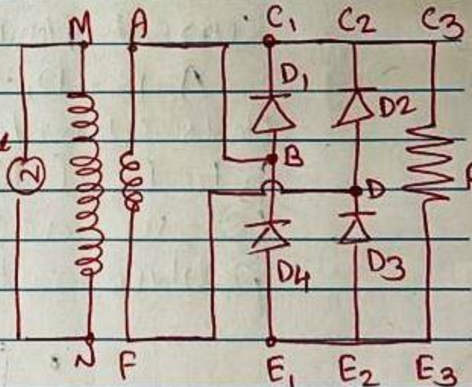
* Bridge Rectifier

- Here four diodes are used in a bridge form as shown in Fig. 1



OR

fig 1 (a)



$C = C_1 = C_2 = C_3$

$E = E_1 = E_2 = E_3$

fig 1 (b)

(20)

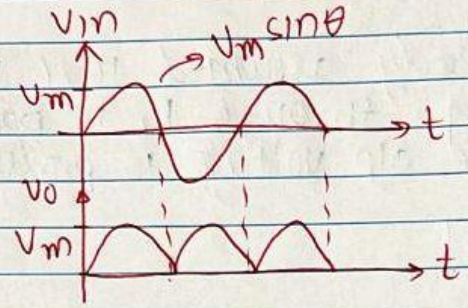


fig ② shows input, output waveforms.

Working (With Reference to fig 1b)

1) for Positive Half Cycle

- M is positive w.r.to N
- A is positive w.r.to F
- Thus B is positive w.r.to D.
- Due to this diodes D_1 & D_3 becomes forward biased and diode D_2 & D_4 becomes reverse biased.
- Current flows through the circuit as -

  A-B-D_1-C_1-C_2-C_3-R_L-E_3-E_2-D_3-D-F.
- Thus current flowing through R_L is from C_3 to E_3 ($R_L \downarrow$)
- Let us assume that this current produces positive output voltage

$$\therefore \boxed{V_0 = +ve}$$

2) for Negative Half Cycle

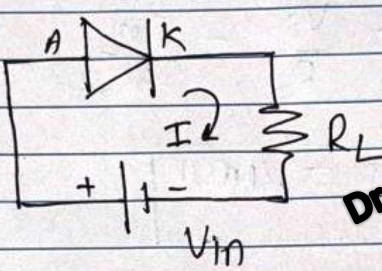
- M is negative, N is positive
- A is negative, F is positive
- B is negative, D is positive
- Due to this diodes D_2 & D_4 becomes forward biased whereas diodes D_1 & D_3 reverse biased.
- Current flows through the circuit as -
  F - D - D_2 - C_2 - C_3 - R_L - E_3 - E_2 - E_1 - D_4 - B - A
- Thus current flowing through R_L is from C_3 to E_3 ($R_L\downarrow$)
- As the current flowing through $R_L\downarrow$ produces positive o/p voltage, we get,

$$\boxed{V_O = +Ve}$$

- Thus, here both the input half cycles are converted into positive so as to get positive output voltage.

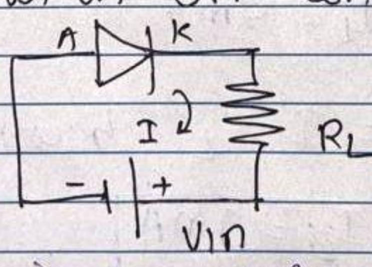
* Diode as a Switch

Case 1) Switch ON condition



- Here, diode is in forward bias condition & turns ON.
- Current flows through circuit & we get output across load.
- This is a swith 'ON' condition.

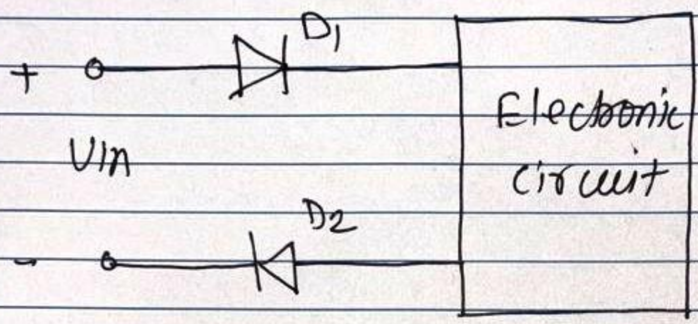
Case 2) Switch OFF Condition



- Here diode is in reverse bias condition & remain in cut off state.
- There is no current flowing through circuit & thus we have zero o/p voltage
- This is switch 'Off' condition.

* Diode as a protection device

- It is basically used for the protection of circuit from supply reversal. as shown below

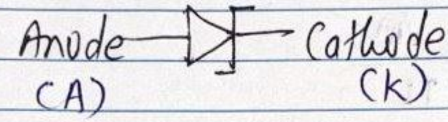


- When the supply voltage is connected properly, both the diodes becomes forward bias.
- This will source the circuit.

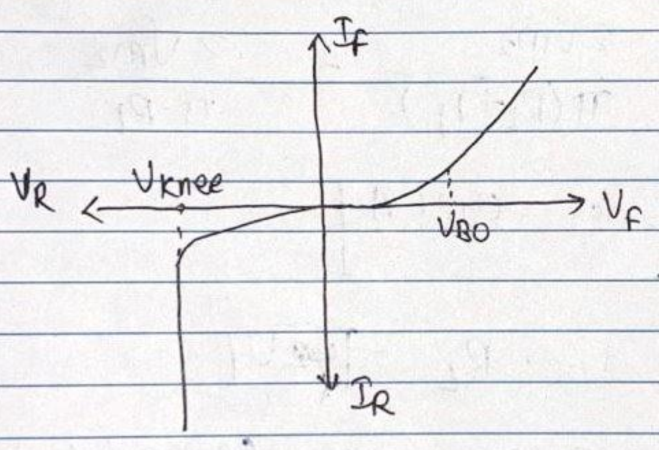
- When supply voltage gets reversed, both diodes becomes reverse biased
- Thus protecting an electronic circuit.

* Zener Diode

- It is a special purpose diode.
- It operates in reverse bias condition.
- It is heavily doped to reduce reverse breakdown voltage
- The symbol of zener voltage is—



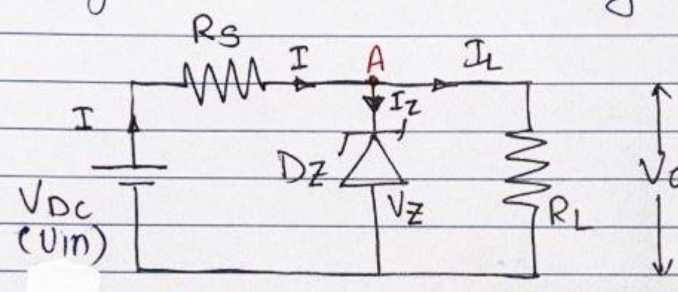
- Zener diode has a sharp breakdown in R/B as shown in V-I characteristics



- for F/B condition, it behaves same as an ordinary diode.

* Zener Diode as a Voltage Regulator

- Regulator is a device which keeps output voltage constant irrespective of change in input voltage and or load current.
- Here zener diode is used as a vtg regulater as shown in fig.



- As shown in figure, zener diode is connected across the load resistance.
- The series resistance R_s is used for current limiting purpose.
- As load is connected across zener,

$$\boxed{V_Z = V_L = V_0 = \text{constant}}$$

- The working of the circuit is explained with the help of two conditions.

Case 1) Vin is fixed & Variable R_L

- (As R_L is variable, so as I_L).

- Here, so as to keep zener in ON state in reverse bias condition, one has to get (R_{Lmin} & I_{Lmax}) and (I_{Lmin} & R_{Lmax})

- for R_{Lmin} & I_{Lmax}

- when R_L is minimum, I_L is maximum.

$$\therefore I_L = \frac{V_0}{R_L}$$

- But $V_0 = V_Z = V_L$ (from fig)

$$\therefore V_0 = \frac{V_{in} \cdot R_L}{R_S + R_L}$$

$$\therefore R_L = \frac{R_S \cdot V_Z}{V_{in} - V_Z} \quad (\because V_0 = V_Z)$$

$$\therefore \boxed{R_{Lmin} = \frac{R_S \cdot V_Z}{V_{in} - V_Z}}$$

But $I_L = \frac{V_0}{R_L} = \frac{V_Z}{R_L}$ $\therefore \boxed{I_{Lmax} = \frac{V_Z}{R_{Lmin}}}$

- for $I_{L min}$ & $R_{L max}$

from fig. Apply KCL at A

$\therefore I = I_Z + I_L$

$\therefore I_Z = I - I_L$

& $I_L = I - I_Z$

$\therefore \boxed{I_{L min} = I - I_{Z max}}$

& we have

$V_0 = V_Z = V_L = I_L \cdot R_L$

$\therefore \boxed{R_{L max} = \frac{V_Z}{I_{L min}}}$

Case 2) R_L is fixed & V_{in} is variable

(As R_L is fixed, so as I_L)

- Here, to turn on zener, one has to get minimum & maximum value of V_{in}.

- for $V_{in(min)}$

we have,

$$V_0 = V_Z = \frac{V_{in} \cdot R_L}{R_S + R_L}$$

$$\therefore \boxed{V_{in(min)} = \frac{(R_S + R_L) \cdot V_Z}{R_L}}$$

- for $V_{in(max)}$

we have, $I = I_Z + I_L$.

OR.

$$I_{max} = I_{Zmax} + I_L$$

Apply KVL to i/p loop, we get

$$V_{in} = I \cdot R_S + V_Z \quad \therefore \boxed{V_{in(max)} = I_{max} \cdot R_S + V_Z}$$